

Code-Layout, Readability and Reusability

Date	June 2026
Team ID	6a3a555ae5895b2c9759029d
Project Name	EduGenie - Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks	
1	Consistent Indentation	Uniform spacing/tabs used throughout the code			
2	Proper File Structure	Files and folders are logically organized			
3	Meaningful Variable Names	Variables reflect their purpose clearly			
	<code>call_gemini_with_retry()</code>	Python (utils/gemini_service.py)	Used indirectly by every <code>get_*</code> () response builder in <code>gemini_service.py</code>	High	
4	Function	<code>log_query() / log_ai_response()</code>	Python (utils/db_manager.py)	Called from every router (qna, explanation, quiz, summary, learning_path) and main.py	High
5	Code Co	<code>get_base_context()</code>	Python (main.py)	Shared by all HTML page routes to pass <code>gemini_configured</code> to templates	Medium
6	Modular	<code>init_local_t5()</code>	Python (explanation_module.py)	Lazily loaded T5 pipeline, reusable for future local-model features	Medium
	<code>clean_json_block()</code>	Python (quiz_module.py)	Strips markdown json fences from LLM output before JSON parsing	Medium	
7	No Redundant Code	Duplicate or unused code is removed			
8	Error Handling	Exceptions and errors are handled gracefully			

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1				

2				
3				
4				
5				

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure		
Readability		
Reusability		
Documentation / Comments		
Overall Score		

call_gemini_with_retry()	Python (utils/gemini_service.py)	Used indirectly by every get_*(*) response builder in gemini_service.py	High
log_query() / log_ai_response()	Python (utils/db_manager.py)	Called from every router (qna, explanation, quiz, summary, learning_path) and main.py	High
get_base_context()	Python (main.py)	Shared by all HTML page routes to pass gemini_configured to templates	Medium
init_local_t5()	Python (explanation_module.py)	Lazily loaded T5 pipeline, reusable for future local-model features	Medium
clean_json_block()	Python (quiz_module.py)	Strips markdown json fences from LLM output before JSON parsing	Medium